3. Identification of Causal Effects

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Identification of Causal Effects* (Prof. Sanghack Lee, GSDS SNU)

The central question is how an SCM defines the distribution of a population under an intervention, and when that interventional distribution can be computed from the causal diagram and the observational distribution alone.

These notes consolidate repeated slides, write out the derivations of the truncated factorization and of adjustment by direct parents (including the optional multi-treatment theorem with full proof), and add a small non-identifiability example. Physical PDF page numbers are listed in Section 6.

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1 Structural causal models, revisited

Purpose. Recall the framework and introduce the do(·) operator.

1.1 Observation versus intervention

Reality is an unknown black-box system. We learn about it in two ways: *observation* (observational studies, passive records) and *intervention* (clinical trials, experiments). Questions such as drug efficacy or the effect of a hypothetical public policy concern the system *under intervention*, even when only observational data are available.

An SCM $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$ (unobserved variables, observed variables, mechanisms for $\mathbf{V}$, distribution of $\mathbf{U}$) is a mathematical understanding of the unknown reality. $\mathbf{V}$ are measured, $\mathbf{U}$ are unknown, and each measured variable is determined by a function of some measured and some unmeasured variables, $v \leftarrow f(\cdot)$. $\mathbf{U}$ can be viewed as what each unit actually is; $\mathbf{U}$ determines $\mathbf{V}$.

The SCM induces a causal diagram $\mathcal{G}$ and a whole family of distributions:

$$\mathcal{M} \longmapsto \mathcal{G},$$
$$\mathcal{M} \longmapsto \{P,\ P_{x_1},\ P_{x_2},\ P_{x_2,x_4}, \dots\}$$

corresponding to $\text{do}(\emptyset), \text{do}(x_1), \text{do}(x_2), \text{do}(x_2, x_4), \dots$

Layer 1 (observational). As derived in Lecture 2,

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{v}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \dots, v_{i-1}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).$$

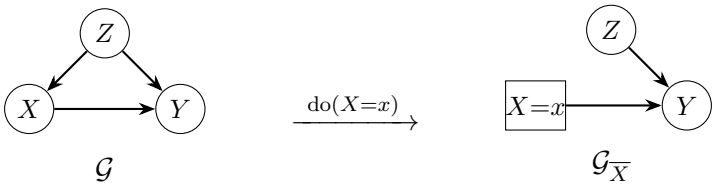
The model also induces Layer 2 (interventional) and Layer 3 (counterfactual) distributions.

Causal diagram. Nodes for $\mathbf{V}$; a directed edge $B \to V_i$ when B is an argument of f_i (causal relationship); a bidirected edge $V_i \leftrightarrow V_j$ when $\mathbf{U}_i$ and $\mathbf{U}_j$ correlate (unobserved confounding).

1.2 The do-operator

Definition 1.1 (Intervention)**.** Given $\mathcal{M}$, fixing $X \in \mathbf{V}$ to a constant x is written $\text{do}(X = x)$. It transforms $\mathcal{M}$ into the *submodel* $\mathcal{M}_x$, i.e. $\mathcal{M}$ with $f_X \in \mathcal{F}$ replaced by the constant x.

Graphically, the diagram of $\mathcal{M}_x$ is $\mathcal{G}_{\overline{X}}$: $\mathcal{G}$ with all edges *into* X removed.



$\mathcal{G}$ $\qquad$ $\mathcal{G}_{\overline{X}}$

2 The second layer: causal effects

Purpose. Define causal effects formally and compute them in Markovian models.

2.1 Connecting different worlds

Data come from P; the query concerns P', the distribution after a change. For instance, would fewer people get cancer if smoking were banned? The query $Q = P(\text{Cancer} = \text{true} \mid \text{do}(\text{Smoking} = \text{no}))$ is *not* an aspect of P.

Observation 1. The distribution alone tells us nothing about change; it describes static conditions of a population under a specific regime.
Observation 2. We must represent change: how the population reacts when it undergoes a change in regime.

The challenge. Let Z be age/sex, X an action, W a mediator, Y an outcome, with $Z \to X \to W \to Y$ and $Z \to Y$. In the hypothetical world X is set to x, cutting $Z \to X$. The real world generates $P(z, x, w, y)$; the hypothetical world generates $P(z, w, y \mid \text{do}(x))$. The goal is to know how much Y changes with X if we vary X between two different constants, free of the influence of Z; such variations are called *causal effects*. What is the relationship between $P(z, x, w, y)$ and $P(y \mid \text{do}(x))$? Is $P(y \mid \text{do}(x)) = P(y \mid x)$?

Definition 2.1 (Causal effect; Pearl 2000, Def. 3.2.1)**.** Given disjoint sets $\mathbf{X}, \mathbf{Y}$, the causal effect of $\mathbf{X}$ on $\mathbf{Y}$, $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$, is a function from $\mathfrak{X}_{\mathbf{X}}$ to the space of distributions over $\mathbf{Y}$. For each $\mathbf{x}$, $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ is the probability of $\mathbf{Y} = \mathbf{y}$ induced by deleting from the model all equations of the variables in $\mathbf{X}$ and substituting $\mathbf{X} = \mathbf{x}$ in the remaining equations:

$$P(\mathbf{y} \mid \text{do}(\mathbf{x})) = P_{\mathbf{x}}(\mathbf{y}), \quad \text{the probability of } \mathbf{Y} = \mathbf{y} \text{ in } \mathcal{M}_{\mathbf{x}}.$$

2.2 Computing causal effects from observational data (Markovian case)

In the example, the observational and post-interventional distributions factorize as

$$\begin{aligned} P(\mathbf{v}) &= P(z)\, P(x \mid z)\, P(w \mid x)\, P(y \mid w, z), \\ P_x(\mathbf{v} \setminus \{X\}) &= P(z) \cdot \underbrace{1}_{X=x \text{ in } \mathcal{M}_x} \cdot P(w \mid x)\, P(y \mid w, z). \end{aligned}$$

All other mechanisms are untouched by the intervention (*modularity*), so their conditional distributions are the same in both worlds. Hence

$$P(y \mid \text{do}(x)) = \sum_{z,w} P(z)P(w \mid x)P(y \mid w, z) \;\neq\; \sum_{z,w} P(z \mid x)P(w \mid x)P(y \mid w, z) = P(y \mid x)$$

in general: conditioning weights z by $P(z \mid x)$, intervening by $P(z)$.

Example 2.2 (Sprinkler: seeing versus doing)**.** Season $\to$ Sprinkler, Season $\to$ Rain, Sprinkler $\to$ Wet $\leftarrow$ Rain, Wet $\to$ Slippery.

$$\begin{aligned} Q_1 = P(\text{wt} \mid \text{Sp} = \text{on}) &= \frac{\sum_{\text{sn,rn}} P(\text{sn})P(\text{Sp} = \text{on} \mid \text{sn})P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})}{\sum_{\text{sn}} P(\text{Sp} = \text{on} \mid \text{sn})P(\text{sn})}, \\ Q_2 = P(\text{wt} \mid \text{do}(\text{Sp} = \text{on})) &= \frac{\sum_{\text{sn,rn}} P(\text{sn})\, P'(\text{Sp} = \text{on})\, P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})}{\sum_{\text{sn}} P'(\text{Sp} = \text{on})P(\text{sn})} \\ &= \sum_{\text{sn,rn}} P(\text{sn})P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn}), \end{aligned}$$

where $P'(\text{Sp} = \text{on}) = 1$ is the interventional probability. Seeing the sprinkler on is evidence about the season and therefore about rain; turning it on is not.

3 Truncated factorization

Purpose. State the general formula for interventional distributions in Markovian models and interpret it as reweighting.

Theorem 3.1 (Truncated factorization; manipulation theorem, g-computation)**.** *In a Markovian model, the distribution generated by* $\mathrm{do}(\mathbf{X} = \mathbf{x})$ *is*

$$P(\mathbf{v} \mid \mathrm{do}(\mathbf{x})) = \begin{cases} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i) & \text{if } \mathbf{v} \text{ is consistent with } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases}$$

Derivation. $\mathcal{M}_{\mathbf{x}}$ is again Markovian, with the same mechanisms f_i and $P(u_i)$ for $V_i \notin \mathbf{X}$ and constant mechanisms for $\mathbf{X}$. Its Bayesian factorization (Lecture 2) is

$$\prod_{V_i \notin \mathbf{X}} P_{\mathbf{x}}(v_i \mid \mathbf{pa}_i) \prod_{X \in \mathbf{X}} \mathbb{1}[x \text{ agrees with } \mathbf{x}].$$

Since f_i and U_i are unchanged and $U_i \perp\!\!\!\perp Pa_i$ in both models,

$$P_{\mathbf{x}}(v_i \mid \mathbf{pa}_i) = \sum_{u_i} \mathbb{1}[f_i(\mathbf{pa}_i, u_i) = v_i] P(u_i) = P(v_i \mid \mathbf{pa}_i)$$

whenever the latter is defined (positivity). □

Reweighting form. Since $P(\mathbf{v}) = \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i) \cdot \prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X)$,

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \frac{P(\mathbf{v})}{\prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X)}, \qquad \mathbf{v}' = \mathbf{v} \setminus \mathbf{x}.$$

For a singleton X, with $\mathbf{v}'' = \mathbf{v} \setminus (\mathbf{pa}_X \cup \{x\})$,

$$P(\mathbf{v}' \mid \mathrm{do}(x)) = \frac{P(\mathbf{v})}{P(x \mid \mathbf{pa}_X)} = \frac{P(\mathbf{v}) P(\mathbf{pa}_X)}{P(x, \mathbf{pa}_X)} = P(\mathbf{v}'' \mid x, \mathbf{pa}_X)\, P(\mathbf{pa}_X).$$

The passage from observational to interventional distribution is a *reweighting*: units with $X = x$ are up-weighted by $1/P(x \mid \mathbf{pa}_X)$ (the origin of inverse probability weighting, Lecture 10).

4 The identification problem

Purpose. Make precise when a causal query is determined by what we can observe and assume.

Definition 4.1 (Identifiability; Pearl 2000, Def. 3.2.2)**.** The causal effect of $\mathbf{X}$ on $\mathbf{Y}$ is *identifiable* from $\mathcal{G}$ if $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ can be computed uniquely from any positive distribution over the observed variables; that is, for every pair of models $\mathcal{M}_1, \mathcal{M}_2$ inducing $\mathcal{G}$,

$$P^{\mathcal{M}_1}(\mathbf{v}) = P^{\mathcal{M}_2}(\mathbf{v}) > 0 \implies P^{\mathcal{M}_1}(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = P^{\mathcal{M}_2}(\mathbf{y} \mid \mathrm{do}(\mathbf{x})).$$

Picture. The true SCM $\mathcal{M}$ is unobserved; it induces the causal graph $\mathcal{G}$ (assumed known), the observational distribution $P(\mathbf{v})$ (observed) and the experimental distribution $P(\mathbf{v} \mid \mathrm{do}(\mathbf{x}))$ (unobserved). Among all models, consider those inducing $P(\mathbf{V})$ and those inducing $\mathcal{G}$. The query is identifiable iff it is constant over the intersection of these two sets: no matter the form of $\mathcal{M}_1, \mathcal{M}_2$, all models agreeing on $\mathcal{G}$ and $P(\mathbf{v})$ agree on $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$. If two models in the intersection disagree, the query is not identifiable.

Example 4.2 (A non-identifiable effect)**.** Let $\mathcal{G}$ be $X \to Y$ with $X \leftrightarrow Y$, binary variables, $U \sim \mathrm{Bern}(1/2)$ shared. $\mathcal{M}_1$: $X \leftarrow U$, $Y \leftarrow U$ (no effect). $\mathcal{M}_2$: $X \leftarrow U$, $Y \leftarrow X$. Both give $P(X = Y = 0) = P(X = Y = 1) = 1/2$, yet $P^{\mathcal{M}_1}(Y = 1 \mid \mathrm{do}(X = 1)) = 1/2$ and $P^{\mathcal{M}_2}(Y = 1 \mid \mathrm{do}(X = 1)) = 1$. (Both are compatible with $\mathcal{G}$, since a graph only permits, never requires, dependence.)

4.1 Identification in Markovian models

Theorem 4.3. *Given the causal diagram $\mathcal{G}$ of any Markovian model (all relevant variables measured), $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable for all disjoint $\mathbf{X}, \mathbf{Y}$, and*

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i),$$
$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{v}' \setminus \mathbf{y}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i).$$

The right-hand side involves only conditionals of $P(\mathbf{v})$, which are fixed by the observational distribution under positivity.

4.2 Adjustment by direct parents (singleton)

Theorem 4.4. *In a Markovian model where only a subset of variables is measured, $Q = P(\mathbf{y} \mid \mathrm{do}(x))$ is identifiable whenever X, $\mathbf{Y}$ and Pa_X are measured, and*

$$P(\mathbf{y} \mid \mathrm{do}(x)) = \sum_{\mathbf{pa}_X} P(\mathbf{y} \mid x, \mathbf{pa}_X) P(\mathbf{pa}_X).$$

Proof. From the singleton reweighting identity, $P(\mathbf{v}' \mid \mathrm{do}(x)) = P(\mathbf{v}'' \mid x, \mathbf{pa}_X) P(\mathbf{pa}_X)$; summing over $\mathbf{v}'' \setminus \mathbf{y}$ gives the claim. Every unmeasured variable is summed out inside $P(\cdot \mid x, \mathbf{pa}_X)$, so only measured variables remain. □

4.3 Adjustment by direct parents for a set of treatments (optional)

For $\mathbf{X} = \{X_1, \ldots, X_k\}$ adjustment by parents is not automatic: a parent of X_j may be a descendant of an earlier X_i. Notation:

$Pa_{\mathbf{X}} = \bigcup_{X \in \mathbf{X}} Pa_X$	parents of $\mathbf{X}$ (may intersect $\mathbf{X}$),
$Pa^{-}_{\mathbf{W}} = Pa_{\mathbf{W}} \setminus \mathbf{W}$	parents of $\mathbf{W}$ excluding $\mathbf{W}$,
$\mathbf{X}_{<i} = \{X_1, \ldots, X_{i-1}\}$, $\mathbf{X}_{\le i}$	predecessors in a topological order,
$Pa'_{X_i} = Pa_{X_i} \setminus (\mathbf{X}_{<i} \cup Pa^{-}_{\mathbf{X}_{<i}})$	"new" parents of X_i.

Theorem 4.5. *Let $\mathbf{X}$ be topologically ordered. If no element of Pa'_{X_j} is a descendant of X_i for every $i < j$, then*

$$\prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X) = P(\mathbf{x} \mid \mathbf{pa}^{-}_{\mathbf{X}}),$$

hence

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{pa}^{-}_{\mathbf{X}}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}^{-}_{\mathbf{X}}) P(\mathbf{pa}^{-}_{\mathbf{X}}).$$

Proof. By induction we show $\prod_{X \in \mathbf{X}_{\le i}} P(x \mid \mathbf{pa}_X) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}})$.

Base case. $P(x_1 \mid \mathbf{pa}_{X_1}) = P(x_1 \mid \mathbf{pa}_{X_1} \setminus \{X_1\}) = P(x_1 \mid \mathbf{pa}^{-}_{X_1}) = P(\mathbf{x}_{\le 1} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le 1}})$.

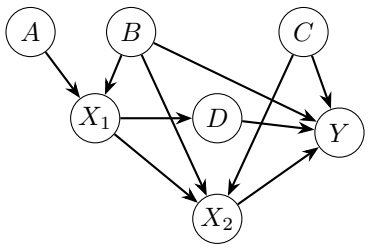
Step. Assume the claim for $i - 1$. It suffices to show $P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) P(x_i \mid \mathbf{pa}_{X_i}) = P(\mathbf{x}_{\le i} \mid$

$\mathbf{pa}^{-}_{\mathbf{X}_{\le i}}$). Note $Pa^{-}_{\mathbf{X}_{\le i}} = Pa^{-}_{\mathbf{X}_{<i}} \cup Pa'_{X_i}$. Then

$$
\begin{aligned}
&P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) P(x_i \mid \mathbf{pa}_{X_i}) \\
&= P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) P(x_i \mid \mathbf{pa}'_{X_i}, \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) && \text{Markov condition} \\
&= P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) P(x_i \mid \mathbf{pa}'_{X_i}, \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) \frac{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \text{multiply by 1} \\
&= \frac{P(\mathbf{pa}'_{X_i}, \mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \text{chain rule, } \{X_i\} \cup \mathbf{X}_{<i} = \mathbf{X}_{\le i} \\
&= \frac{P(\mathbf{pa}'_{X_i}, \mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \mathbf{X}_{<i} \perp\!\!\!\perp Pa'_{X_i} \mid Pa^{-}_{\mathbf{X}_{<i}} \\
&= P(\mathbf{x}_{\le i} \mid \mathbf{pa}'_{X_i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}}).
\end{aligned}
$$

The first line replaces the parents of X_i by a superset of non-descendants (local Markov: $\mathbf{X}_{<i}$ and $Pa^{-}_{\mathbf{X}_{<i}}$ are non-descendants of X_i since they precede it or are parents of predecessors). The key independence holds by d-separation: Pa'_{X_i} are non-descendants of every X_j, $j < i$, so every path from $\mathbf{X}_{<i}$ to Pa'_{X_i} either enters X_j through a parent (blocked by $Pa^{-}_{\mathbf{X}_{<i}}$) or leaves X_j along a directed edge, in which case it must meet a collider that is not an ancestor of the conditioning set. □

Example 4.6. Graph: $A \to X_1$, $B \to X_1$, $B \to X_2$, $B \to Y$, $C \to X_2$, $C \to Y$, $X_1 \to D$, $X_1 \to X_2$, $D \to Y$, $X_2 \to Y$.



Here $Pa_{X_1} = \{A, B\}$, $Pa_{X_2} = \{X_1, B, C\}$, $Pa^{-}_{\mathbf{X}} = \{A, B, C\}$ and $Pa'_{X_2} = \{C\}$, which is not a descendant of X_1. First, $P(y \mid \mathrm{do}(\mathbf{x})) = \sum P(\mathbf{v}) / [P(x_1 \mid a, b) P(x_2 \mid x_1, b, c)]$. Focusing on the denominator,

$$
\begin{aligned}
P(x_1 \mid a, b) P(x_2 \mid x_1, b, c) &= P(x_1 \mid a, b) P(x_2 \mid \underbrace{x_1}_{\mathbf{X}_{<2}}, \underbrace{a, b}_{Pa^{-}_{\mathbf{X}_{<2}}}, \underbrace{c}_{Pa'_{X_2}}) \\
&= P(x_1 \mid a, b) P(x_2 \mid x_1, a, b, c) \frac{P(c \mid x_1, a, b)}{P(c \mid x_1, a, b)} \\
&= \frac{P(x_1, x_2, c \mid a, b)}{P(c \mid x_1, a, b)} = \frac{P(x_1, x_2, c \mid a, b)}{P(c \mid a, b)} = P(x_1, x_2 \mid a, b, c) = P(\mathbf{x} \mid \mathbf{pa}^{-}_{\mathbf{X}}),
\end{aligned}
$$

so $P(y \mid \mathrm{do}(x_1, x_2)) = \sum_{a,b,c} P(y \mid x_1, x_2, a, b, c) P(a, b, c)$. D (a descendant of X_1) is correctly *not* adjusted for.

4.4 When a parent is latent

In the sprinkler model, suppose Season is unobserved. Is Q_2 still computable?

$$\begin{aligned} Q_2 &= \sum_{\text{sn,rn}} P(\text{sn})P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn}) \\ &= \sum_{\text{sn,rn}} P(\text{rn}, \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn}) \\ &= \sum_{\text{rn}} P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})P(\text{rn}). \end{aligned}$$

Yes: adjusting for Rain instead of the latent parent Season suffices. Adjustment sets other than the parents can work; characterizing them is the subject of the back-door criterion (Lecture 4).

5 Summary

- Causal effects $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ are defined through submodels (deleting and substituting equations).
- In Markovian models every causal effect is identifiable via the truncated factorization.
- Adjustment by direct parents gives a practical formula: condition on the parents of the treatment and average over their marginal distribution.
- The identification problem asks whether $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ is uniquely determined by $\mathcal{G}$ and $P(\mathbf{v})$.
- With latent variables identification becomes non-trivial; this continues with the back-door criterion and beyond.

6 Source coverage

The source deck has 72 physical pages (31 numbered slides with build-ups).

Topic	Physical pages
SCM review, distributions, diagrams, do-operator	1–18
Layer 2: causal effects, definitions, examples	19–31
Truncated factorization	32–38
Identification problem and Venn picture	39–58
Markovian identification, parent adjustment (incl. optional proof)	59–65
Latent Season example, summary, references	66–72